

CamLoop4D: Controllable 4D Scene Generation by Closing the Camera Loop between Generation and Reconstruction

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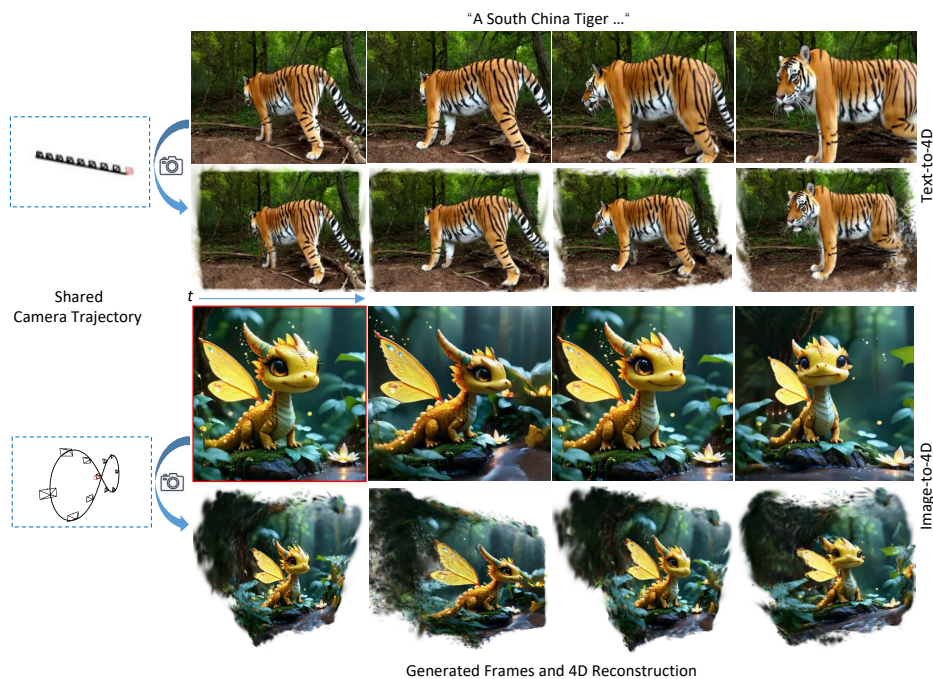


Figure 1: CamLoop4D generates **controllable** 4D scenes from a text prompt or a single image under user-specified camera trajectories; **the red border indicates the input conditioning image for the Image-to-4D task**. By sharing the same Plücker trajectory tensor between a camera-conditioned video diffusion model and a 4D Gaussian reconstructor, our co-design framework keeps the two stages geometrically aligned without pose re-estimation, producing temporally coherent 4D content on a single RTX 3090 in ~ 40 minutes.

Abstract

Generating dynamic 3D scenes with coherent geometry and motion from a text prompt or a single image, under user-specified camera trajectories, is a long-standing goal in computer graphics content creation. Existing pipelines cascade a video generator and a 4D reconstructor through an error-prone pose-estimation step, where estimation errors compound with generation artifacts. We present CamLoop4D, a generation-reconstruction co-design framework that closes this loop: a camera-conditioned video diffusion model and a 4D Gaussian reconstructor operate on the same Plücker trajectory tensor, removing pose re-estimation by construction, and the interface is generator-agnostic. For reconstruction, we introduce hybrid motion bases that combine six fixed $\mathfrak{se}(3)$ generators encoding rigid-body priors with learnable non-rigid bases, reducing motion-bank storage from $\mathcal{O}(B \cdot F)$ to $\mathcal{O}(B)$ while improving fidelity; **under the matched-input protocol (all methods given the same real video and shared poses), the reconstruction module surpasses the strongest monocular baselines**. CamLoop4D achieves the best PSNR/SSIM among all compared methods on the iPhone benchmark and the highest user-preference rates against seven recent baselines in text-to-4D and image-to-4D evaluation, and it generates an 80-frame 960×720 scene in ~ 40 minutes on a single RTX 3090 with real-time 20fps playback, bringing controllable 4D content creation within reach of individual creators.

CCS Concepts

• **Computing methodologies** $\rightarrow$ **Animation**; **Rendering**; **Shape modeling**;

1. Introduction

Dynamic 3D content lies at the heart of modern computer graphics, powering virtual production, games, visual effects, and immersive VR/AR experiences, yet authoring it remains labour-intensive: artists must model geometry, rig motion, and choreograph cameras. A long-standing goal is therefore to generate dynamic 3D scenes (i.e., 4D content) with full spatiotemporal consistency from minimal input, such as a text prompt or a single image, while retaining the camera-level control that graphics workflows demand.

Despite significant advances in static 3D reconstruction [MST*21, KKLD23], synthesizing dynamic 3D scenes with semantic and temporal coherence from a single image or text, guided by user-specified camera trajectories, remains **open**. Existing dynamic reconstruction methods typically require synchronized multi-view videos [FKMW*23, LKLR24, WYF*24] or depth sensors [GWW*19, PBM20, WLPL22, WLHJ*20], and scene-level navigation approaches [YDH*24, YDH*25] offer only step-wise control without fine-grained trajectory conditioning.

Recent camera-controlled video diffusion models [HXG*25, ZLJ*24, HC25, BSS*25, BSQ*25, ZGV*25] and 4D reconstruction via 3D Gaussian splatting [KKLD23] supply the necessary building blocks, but *naively cascading* the two stages creates a critical bottleneck: an intermediate pose-estimation step introduces errors that compound with generation artifacts. Moreover, existing dynamic representations are either fully unconstrained and prone to overfitting, or overly rigid and unable to capture complex deformations.

We observe that the root cause of this bottleneck is a *design mismatch*: the camera parameters that already condition the generative model are discarded and re-estimated before reconstruction. CamLoop4D eliminates this mismatch through **generation-reconstruction co-design**, in which both stages consume the same Plücker trajectory tensor, so the reconstructor reads the specified camera parameters directly instead of recovering them post hoc. We are precise about scope. **Sharing** the trajectory removes the *pose-estimation stage* from the pipeline, **though** it does not, by itself, force the generated frames to obey the specified path perfectly (we quantify the residual generator deviation in Section 3.2.3). To our knowledge no prior open-prompt 4D system reuses the conditioning trajectory in this way. **Our** contribution is to formalise this interface, quantify its benefit, and validate it across generators. Both stages remain frozen and modular, **because** the co-design is an interface contract rather than joint training — precisely what keeps the pipeline practical on consumer hardware and portable to future generators. Our contributions are:

(1) **A shared-Plücker-trajectory interface** (a one-way convention, not joint training or feedback) treating cross-stage trajectory consistency as an explicit design axis. It yields +0.88 dB PSNR over poses from the *static* estimator DUST3R [WLC*24], or a still-positive +0.34 dB against dynamic-aware MonST3R; the generator is used *frozen, off-the-shelf*, so the interface, not a new generator, is the contribution [BSQ*25, HXG*25].

(2) **Hybrid motion bases for persistent 3D Gaussians** that combine six fixed $\mathfrak{se}(3)$ generators encoding rigid-body priors with learnable bases. With per-Gaussian per-frame coefficients the six generators already span the instantaneous motion space; the train-

able bases act as a shared, scene-adapted low-rank prior that *conditions and regularises* the optimisation. This raises reconstruction fidelity by +0.64 dB PSNR over a fully-learnable basis set, and under matched real-video inputs improves on SoM by +0.52 dB and on the strongest baseline MoSca by a small but significant aggregate margin (+0.19 dB PSNR), at the cost of a larger per-frame coefficient tensor: a deliberate expressiveness-for-memory trade we quantify in Section 3.2.2. The advantage replicates on a second dynamic benchmark and is independent of the generative front-end.

(3) **A practical single-GPU system** for text/image-to-4D generation featuring unified foreground-background modelling and Gaussian downsampling. CamLoop4D completes an 80-frame 960×720 sequence in ~ 40 minutes on a single RTX 3090 while rendering at 20 fps, over an order of magnitude cheaper than methods requiring multi-GPU clusters [WGP*25], bringing high-quality 4D content creation within reach of individual artists and small studios.

2. Related Work

Video Diffusion Models and Camera Control. Latent diffusion models [RBL*22, SME22] have been extended from images to video [HSG*22, HCS*22, HYZ*22, HDZ*22, CXH*23, ZWZ*23, BPH*24], with adapter-based approaches such as AnimateDiff [GYR*24b] enabling motion priors to be grafted onto pre-trained image generators. Building on these backbones, a rapidly growing line of work injects explicit camera control: CameraCtrl [HXG*25], CamI2V [ZLJ*24], VD3D [BSS*25], AC3D [BSQ*25], and Stable Virtual Camera [ZGV*25] encode camera trajectories as Plücker embeddings and inject the extracted multi-scale features into the (temporal) attention layers of video diffusion models, while training-free alternatives manipulate latents directly [HC25]. These models enable controllable camera motion for filmmaking and VR/AR applications [CWX*23, GYR*24a, Hu24, KMT*23, XZL*24]. However, they operate purely in pixel space and lack an explicit 3D representation, **so** their outputs cannot be re-rendered from novel viewpoints, edited at the object level, or played back in real time. Crucially, when such videos are fed to a downstream reconstructor, the camera parameters that conditioned the generation are typically discarded and re-estimated, a redundancy that CamLoop4D eliminates.

Dynamic Scene Reconstruction. Classical non-rigid reconstruction relies on RGB-D sensors [NFS15, DKD*16, IZN*16, ZNI*14, BZTN20] or hand-designed priors [KDL17, RVCK16, RYA14], while multi-view rigs enable high-quality volumetric video [BFO*20, LSZ*22, IRG*23, WCS*22]. Neural radiance fields [MST*21] spurred a family of dynamic extensions, including deformation-field methods [PSB*21, PSH*21, PCPMMN21], space-time radiance fields [XHKK21, LNSW21, WELG21, FKMW*23, CJ23, SCL*23], and image-based rendering for long videos [LWC*23]; monocular settings additionally exploit consistent video depth [YKG*20, LHS*20, KRH21, ZCT*21]. 3D Gaussian Splatting [KKLD23] brought real-time rendering to this problem, with dynamic variants based on per-frame Gaussians [LKLR24], deformation networks [WYF*24, YGZ*24], 4D primitives [YYPZ24, DWD*24], and compact mo-

tion bases [KLD24, LCLX24]. Most recently, SoM [WYG*25] and MoSca [LWH*24] substantially improved monocular reconstruction by incorporating auxiliary signals from large vision models, e.g., segmentation [KMR*23], depth [YKH*24, PYS*24], and point tracking [DYV*23, KRG*24]. While these methods perform well on captured videos, they cannot generate 4D content from text or images, and they invariably depend on pose estimation [TD21, WLC*24] whose errors propagate into the reconstruction. CamLoop4D bridges this gap by sharing the Plücker trajectory tensor across generation and reconstruction, removing the pose re-estimation *stage* from the pipeline (without claiming the generated frames follow the path perfectly; cf. Section 3.2.3); its hybrid motion bases further advance reconstruction quality itself (+0.52 dB over SoM under matched inputs).

3D and 4D Generation. Early 3D generation targeted static objects [ADMG18, PFS*19, GSW*22] or static scenes [CWL23, LLM*23], with score distillation [PBJM23] enabling text-driven synthesis. Subsequent work added temporal deformation for object-level 4D generation [SSP*23, ZLN*24, RPT*23, ZYX*23, YXW*23, RXM*24, SGW*24, PYZZ26], which, however, produces isolated objects without scene context or camera-path control. Scene-level navigation approaches [YDH*24, YDH*25] offer step-wise exploration but lack fine-grained trajectory conditioning. Recent scene-level 4D generators include DimensionX [SCL*25], GenXD [ZLL*25], 4Real [YWZ*24], CAT4D [WGP*25], and Free4D [LHC*25]. CAT4D achieves high quality but requires 8×A100 GPUs, whereas Free4D is tuning-free but re-estimates camera poses instead of sharing them with the reconstruction stage. In contrast, CamLoop4D identifies *cross-stage trajectory consistency* as the key missing axis. A frozen video model is coupled with the reconstructor through the same Plücker trajectory, closing the loop that cascaded pipelines leave open, and the purely Plücker-level interface is validated with two generators and extensible to future camera-conditioned models.

3. Method

We target 4D scene generation from open natural language prompts or single images. Figure 2 illustrates the overall pipeline of CamLoop4D, comprising two tightly coupled stages: camera-controlled video generation (Section 3.1) and 4D scene reconstruction (Section 3.2). The central design principle, *generation-reconstruction co-design*, manifests in three ways: (i) both stages consume the same Plücker trajectory tensor, removing the pose re-estimation *stage* from the pipeline (we quantify the residual generator deviation in Section 3.2.3); (ii) the reconstruction stage introduces hybrid motion bases that combine rigid-body priors with learnable deformation, providing a structured yet expressive 4D representation; and (iii) the interface is generator-agnostic, depending only on the Plücker tensor and not on any specific generator architecture. This eliminates the fragile pose-estimation bridge of cascaded pipelines while running entirely on a single consumer GPU.

3.1. Camera-Controlled Video Generation

Camera Representation. Following CameraCtrl [HXG*25], we adopt Plücker embeddings [SRF*21] to encode camera trajectories.

For each pixel (u, v) , the Plücker embedding is:

$$p_{u,v} = [o \times d_{u,v}; d_{u,v}] \in \mathbb{R}^6, \quad (1)$$

where $[\cdot; \cdot]$ denotes *concatenation* of the two 3-vectors (not an inner product), $o \in \mathbb{R}^3$ is the camera center in world coordinates, and $d_{u,v}$ is the normalized ray direction computed from the camera intrinsics $\mathbf{K}$ and extrinsics $[\mathbf{R}|\mathbf{t}]$:

$$d_{u,v} = \text{normalize}(\mathbf{R}\mathbf{K}^{-1}[u, v, 1]^T), \quad (2)$$

with $\mathbf{R} \in \mathbb{SO}(3)$ and $\mathbf{t} \in \mathbb{R}^3$. The full camera trajectory is then represented as a sequence $\{P_i\}_{i=1}^n$ with $P_i \in \mathbb{R}^{6 \times h \times w}$.

Video Generation. We leverage the pre-trained camera-controlled video diffusion model AC3D [BSQ*25], which builds upon CogVideoX [YTZ*25]. AC3D uses a learned encoder to extract multi-scale features from the Plücker embeddings and injects them into the temporal attention layers of the diffusion model. The standard latent diffusion objective is:

$$L(\theta) \triangleq \mathbb{E}_{z,c,\varepsilon \sim \mathcal{N}(0,I),t} [\|\varepsilon - \varepsilon_\theta(z_t, c, t)\|_2^2], \quad (3)$$

where $z = \mathcal{E}(x)$ is the latent video encoding, c is the conditioning input (text prompt or reference image), ε_θ is the denoiser, and $z_t = \alpha_t z_0 + \sigma_t \varepsilon$ with $\sigma_t = \sqrt{1 - \alpha_t^2}$ [HJA20].

We use the publicly available AC3D checkpoint without additional fine-tuning; the trajectory tensor $\{P_i\}$ produced here is *directly shared* with the reconstruction stage (Section 3.2.3).

3.2. 4D Scene Reconstruction

3.2.1. Preliminaries

We represent dynamic scenes using persistent 3D Gaussians [KKLD23, WYG*25]. Each Gaussian in the canonical frame t_0 is parameterised by $g_0 = (\mu_0, R_0, s, o, c)$, where $\mu_0 \in \mathbb{R}^3$ is the mean, $R_0 \in \mathbb{SO}(3)$ the rotation, $s \in \mathbb{R}^3$ the scale, o the opacity, and $c \in \mathbb{R}^3$ the colour (s, o, c are time-invariant). Rendering follows the standard splatting pipeline [KKLD23]: each Gaussian is projected onto the image plane via an affine approximation and alpha-blended to produce RGB images and depth maps.

3.2.2. Dynamic Scene Representation

For dynamic scene representation and motion modelling, inspired by SoM [WYG*25], we design a persistent 3D Gaussian representation. Motion is modelled using a set of compact and globally shared hybrid motion bases (each a $\mathfrak{se}(3)$ Lie-algebra element, converted to $\mathbb{SE}(3)$ by the exponential map), with each Gaussian trajectory expressed as a linear combination thereof, enabling efficient representation of complex dynamic behaviours.

To model dynamic 3D scenes, we maintain a set of N canonical 3D Gaussian distributions, whose positions and orientations are transformed over time via rigid transformations to simulate motion. The transformation from the canonical frame t_0 to t is denoted by $T_{0:t} = [R_{0:t}; t_{0:t}] \in \mathbb{SE}(3)$. The pose of each Gaussian at t , given by (μ_t, R_t) is computed as:

$$\begin{aligned} \mu_t &= R_{0:t}\mu_0 + t_{0:t}, \\ R_t &= R_{0:t}R_0, \end{aligned} \quad (4)$$

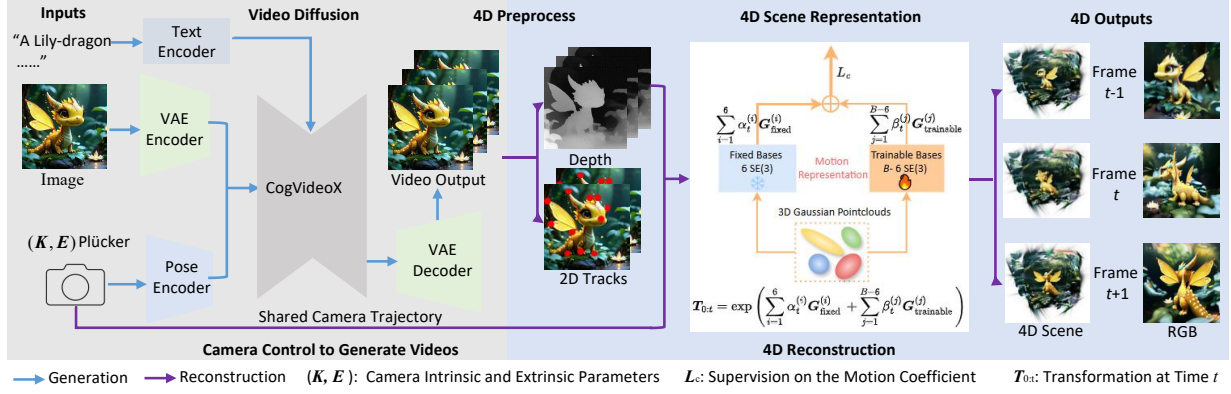


Figure 2: **System Overview.** We first encode a text prompt or single image into a latent spatial representation using a VAE encoder. Concurrently, based on camera intrinsic $\mathbf{K}$ and extrinsic $\mathbf{E}$ parameters, the specified trajectory is encoded using Plücker coordinates and shared by the generation and reconstruction stages, removing pose re-estimation. AC3D [BSQ*25] (Section 3.1) then generates a video sequence with the desired camera motion; off-the-shelf models [DYV*23, YKH*24] extract depth maps and 2D point trajectories, which, with the RGB frames, feed reconstruction. Dynamic scenes are represented by persistent 3D Gaussians whose motion is a linear combination of globally shared, compact hybrid motion bases (each an $\mathfrak{se}(3)$ element mapped to $\mathbb{SE}(3)$ via the exponential map) for efficient modelling of complex dynamics (Section 3.2).

where μ_t and R_t represent the 3D position and rotation of the Gaussian at t .

At any time t , the transformation $T_{0:t} \in \mathbb{SE}(3)$ is computed via the exponential map of a weighted sum of $\mathfrak{se}(3)$ basis elements:

$$T_{0:t} = \exp \left(\sum_{i=1}^6 \alpha_t^{(i)} G_{\text{fixed}}^{(i)} + \sum_{j=1}^{B-6} \beta_t^{(j)} G_{\text{trainable}}^{(j)} \right), \quad (5)$$

where $G_{\text{fixed}}^{(i)} \in \mathfrak{se}(3)$ are the six frozen generators (three translations and three rotations, see Appendix) and $G_{\text{trainable}}^{(j)} \in \mathfrak{se}(3)$ are learnable bases. The coefficients $\alpha_t^{(i)}$ and $\beta_t^{(j)}$ are per-Gaussian and per-frame; we omit the Gaussian index n in Equation (5) for readability (it is made explicit in Equation (6)). The set of motion bases is globally shared across all Gaussians [KLD24]; only the coefficients are per-Gaussian.

Mathematical justification. Since $\mathfrak{se}(3)$ is a vector space, any linear combination $\xi = \sum_b \gamma^{(b)} G^{(b)}$ remains in $\mathfrak{se}(3)$, and $\exp(\xi) \in \mathbb{SE}(3)$ is a valid rigid transformation. Although $\mathbb{SE}(3)$ is non-commutative, the motion coefficient loss (Equation (6)) keeps coefficients small (mean absolute magnitude 0.037 ± 0.021 , 95th percentile 0.082 across all 14 iPhone sequences), ensuring the first-order BCH approximation holds reliably. Each basis is compactly stored as a 6D vector and converted to $\mathbb{SE}(3)$ via the exponential map during the forward pass [WYG*25].

Role of the trainable bases. Two notions are easily conflated. (i) *Expressiveness*: with per-Gaussian *per-frame* coefficients the six fixed generators (which already span $\mathfrak{se}(3)$) already suffice to give every Gaussian an arbitrary rigid transform at every frame, so the trainable bases add no instantaneous degrees of freedom; scene non-rigidity comes from different Gaussians moving differently. (ii) *Conditioning*: the trainable bases act as a shared, scene-adapted low-rank dictionary that couples Gaussians, letting coherent motion be explained by a few globally shared directions rather

than fitting noisy per-Gaussian coefficients independently against imperfect supervision. This is the shared-low-rank-motion prior of compact motion-basis methods [KLD24, LCLX24, WYG*25]; our addition is the fixed rigid scaffold, justified empirically (Section 4.3). Removing the trainable bases lets the per-frame coefficients overfit noisy tracks and collapses PSNR by ~ 5 dB; this is a conditioning failure, not lost expressiveness (multi-seed controls, an alternative-regulariser comparison, a temporal hold-out, and the zero-init/symmetry-breaking details are in the supplement).

Comparison with SoM and storage. SoM [WYG*25] stores time-varying bases $B \in \mathbb{R}^{B \times F \times 6}$ with time-invariant coefficients $C \in \mathbb{R}^{N \times B}$; we move the time dependence into the coefficients, storing static bases $B \in \mathbb{R}^{B \times 6}$ with per-Gaussian per-frame coefficients $C \in \mathbb{R}^{N \times F \times B}$. The basis bank thus shrinks from $\mathcal{O}(B \cdot F)$ to $\mathcal{O}(B)$, but this is *not* a net storage win: the $\mathcal{O}(N \cdot F \cdot B)$ coefficient tensor now dominates ($N \gg F, B$), making our total motion state ≈ 240 MB versus SoM’s ≈ 3 MB (supplement). We make no compactness claim; we trade $\approx 80\times$ larger (but still small, well under the 24 GB budget) motion state for the per-frame expressiveness that, with the rigid scaffold, yields the +0.52 dB gain over SoM under matched inputs (Table 2).

3.2.3. Shared Camera Trajectory Interface

We share the Plücker trajectory $\{P_i\}$ from Section 3.1 directly with the reconstructor, which therefore consumes the specified poses rather than re-estimated ones. This removes the estimation step whose errors dominate cascaded pipelines (it does not force the generated frames to obey the path). Both stages share one convention: the intrinsics $\mathbf{K}$ that condition the generator, a world frame at the canonical frame t_0 , and the metric scale fixed by the specified trajectory; Depth Anything [YKH*24] depth is least-squares shift/scale-aligned per frame and TAPIR [DYV*23] tracks are lifted with the same intrinsics, so no separate scale or frame resolution is needed. **TAPIR queries are sampled uniformly over the**

frame with no foreground restriction, 256 per sequence, and back-projected with the shared intrinsics and aligned depth. The shared low-rank motion prior (Section 3.2.2) absorbs noisy or sparse track supervision, so reconstruction is not sensitive to the exact track density (removing the trainable bases collapses held-out PSNR by ~ 5 dB, Section 4.3; supervision-robustness regimes are analysed in Supp. C.7). As a diagnostic only (computed offline with COLMAP, never fed back), the residual AC3D deviation is small (ATE 0.021 m, angular 1.3°); being COLMAP-based on dynamic content, we read it as an indicative upper bound, not a precise measurement. Sharing yields $+0.88$ dB over DUST3R [WLC*24]-estimated poses (Table 6). Since DUST3R is a weak static estimator (0.142 m ATE), we also re-estimate from the same frames with stronger ones: the gain shrinks to $+0.53$ dB (masked COLMAP) and $+0.34$ dB (dynamic-aware MonST3R [ZHT*24]) but persists, and sharing additionally avoids the estimator’s runtime and failure modes (full table in the supplement). To quantify robustness to generator-induced trajectory inconsistency, we perturb the poses fed to the reconstructor by a controlled ATE (supplement): the loss is < 0.05 dB at AC3D’s residual (≈ 0.02 m), reaches only ≈ 0.8 dB at 0.10 m, and decays sharply only beyond ≈ 0.2 m — well past AC3D’s residual and beyond where estimation error would sit (DUST3R 0.142 m). Residual inconsistencies therefore surface as LPIPS texture smoothing rather than geometric drift. The interface is generator-agnostic: replacing AC3D with CameraCtrl [HXG*25] costs only -0.47 dB (supplement).

3.2.4. Unified Modelling and Downsampling

Unlike SoM [WYG*25], which separates foreground and background, we treat both as co-moving entities under a unified dynamic representation with shared motion bases, better suited for camera-controlled generation where everything moves; instance segmentation [KMR*23] delineates regions within the same basis framework. To prevent excessive Gaussian density near scene boundaries, we downsample dynamic Gaussians by $0.5\times$ during training, improving boundary coherence and reducing memory usage.

3.3. Optimization

Training Details. We use Adam with learning rate 10^{-4} , 1,000 iterations of initial fitting followed by 600 epochs of joint optimisation. We set $B=15$ motion bases (6 fixed + 9 trainable), selected via ablation (Section 4.3), initialise 50,000 Gaussians with adaptive density control [KKLD23], and apply $0.5\times$ downsampling. All experiments run on a single RTX 3090.

Computational Efficiency. CamLoop4D’s end-to-end time on a single RTX 3090 breaks down as: video generation (~ 5 min), preprocessing (~ 5 min), and 4D reconstruction (~ 30 min), totaling ~ 40 min at 20 fps rendering, with a peak GPU memory of ~ 11 GB during optimisation and ~ 3 GB at render time (the ≈ 240 MB motion state is a small part of this). For a hardware-normalised view, CamLoop4D costs ≈ 0.7 GPU-hours (RTX 3090); CAT4D [WGP*25] reports $8\times A100$ for ~ 1 h (≈ 8 A100-GPU-hours), Free4D [LHC*25] $1\times A100$ for ~ 2 h, and SoM [WYG*25] $1\times A6000$ for ~ 1 h. We report these as the methods’ stated operating points, not strict same-hardware benchmarks; even discount-

ing the GPU-class difference, the order-of-magnitude gap to multi-GPU systems is clear.

We introduce additional supervision on the motion coefficients via a motion coefficient loss applied to each Gaussian n at each frame f :

$$L_c := \frac{1}{NF} \sum_{n=1}^N \sum_{f=1}^F \left[(1-\lambda) \sum_{i=1}^6 \left(\alpha_{n,f}^{(i)} \right)^2 + \lambda \sum_{j=1}^{B-6} \left(\beta_{n,f}^{(j)} \right)^2 \right], \quad (6)$$

where $\alpha_{n,f}^{(i)}$ and $\beta_{n,f}^{(j)}$ are the fixed-basis and trainable-basis coefficients of the n -th Gaussian at frame f , N is the number of Gaussians, F the number of frames, and $\lambda=0.8$. The overall training loss is $L = L_{\text{rgb}} + \mu_d L_{\text{depth}} + \mu_t L_{\text{track}} + \mu_c L_c$, where L_{rgb} , L_{depth} , and L_{track} are the photometric, depth, and trajectory supervision terms following SoM [WYG*25], with $\mu_d=0.5$, $\mu_t=0.2$, and $\mu_c=0.01$. The asymmetric weighting ($\lambda=0.8$) more strongly damps the learnable coefficients, encouraging the optimiser to explain motion with rigid generators first.

4. Experiments

We evaluate CamLoop4D on 4D reconstruction (iPhone/DyCheck and NVIDIA Dynamic Scenes), text/image-to-4D generation against seven recent baselines, cross-generator portability, and content-creation applications. Static figures cannot convey temporal coherence; please see the *supplementary video*.

4.1. 4D Reconstruction

Protocol. We use the iPhone dataset [GLT*22] (14 sequences) with PSNR, SSIM [WBSS04], and LPIPS [GLT*22, HZZ*20]. Following the DyCheck protocol these are *held-out novel-view* metrics (the model is fit on the training camera and scored on held-out co-visible cameras), so they test 3D structure, not training-view fit. We report unmasked full-frame metrics over all 14 sequences, so our values are systematically lower than published DyCheck mP-PSNR figures (≈ 19.3 with provided poses, and ≈ 18.8 – 19.0 even with no pose input, for MoSca), which are computed under the co-visibility mask on a 7-sequence multi-camera split; the two are therefore not directly comparable. The *matched-input* comparison is our controlled experiment and the sole basis for architectural claims: all methods receive the same real video with the same DUST3R [WLC*24] poses. The *system-level* comparison instead places each pipeline in its deployment setting (generated video for us, real for baselines); since inputs differ it is deployment context, not a controlled comparison. The *system-level* protocol is detailed in Supp. C.4: AC3D is conditioned on only the *real first frame* and the sequence’s *ground-truth trajectory* at the same frame count and resolution, so the generated and real frames correspond one-to-one and are scored against the real held-out frames *without* any search, warping, or colour matching. The scenes are *not* treated as static, and the metrics are *not* computed on training views.

Matched-input (primary). Our module reaches 16.41 dB / 0.621 / 0.440 (Table 2), $+0.52$ dB over SoM and $+0.18$ dB over MoSca. We are candid that the MoSca margin is small and *non-uniform*: per scene (full table in the supplement) CamLoop4D wins 11 of 14 but

Table 1: Matched-input reconstruction on a *second* benchmark, NVIDIA Dynamic Scenes [YKG*20] (7 monocular sequences), same protocol as the iPhone matched-input experiment. The ordering replicates. Best in **bold**.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
SoM [WYG*25]	21.24	0.71	0.21
MoSca [LWH*24]	21.74	0.73	0.19
Ours (recon. only)	21.94	0.74	0.18

loses the three most-articulated scenes (*block*, *hand*, *mochi-high-five*), where MoSca’s separate foreground/background helps. The per-sequence paired test gives +0.19 dB over MoSca ($t(13)=2.72$, $p=0.009$, one-sided) and +0.52 dB over SoM ($t(13)=6.11$). **Our claim, scoped in Section 5, is a small but significant aggregate gain from the hybrid bases and unified modelling, independent of the generator.** Raw per-run logs and the eval script will be released.

Second benchmark. To show the advantage is not specific to iPhone statistics, we repeat the matched-input protocol (same real video and DUST3R poses for all methods) on the NVIDIA Dynamic Scenes dataset [YKG*20] (7 monocular sequences, Table 1). The ordering replicates: +0.20 dB over MoSca and +0.70 over SoM, with the MoSca margin at the edge of significance at this small sample size ($t(6)=1.87$, $p=0.055$). We read this as corroborating the *direction* of the effect on a second dataset, not as a second strong-significance claim.

System-level (deployment). The system-level comparison is deployment context, not a controlled comparison (protocol in Supp. C.4). From *generated* video CamLoop4D reaches 16.55 dB / 0.62 (Table 2), on par with real-video baselines (MoSca 16.23, SoM 15.89). This is not like-for-like: the rough parity arises because sharing poses avoids the ≈ 0.88 dB DUST3R penalty (Table 6) that offsets the lower fidelity of generated frames. The higher LPIPS (0.48 vs. 0.45) is generated-video texture smoothing, not architecture (our matched-input LPIPS is at parity with MoSca, 0.440 vs. 0.443). The interface is also generator-agnostic: swapping AC3D for CameraCtrl [HXG*25] costs only -0.47 dB (supplement).

Qualitative. In Figure 3, 4D Gaussians and HyperNeRF are blurry with geometric distortion and MoSca/SoM retain view-dependent artifacts, whereas CamLoop4D recovers cleaner geometry and more stable high-frequency detail (e.g. the flower-pot pattern, the rolling arm). It also preserves dynamic layout and object identity under viewpoints *outside* the input trajectory (qualitative novel-view renderings in the supplement; no ground truth exists for extrapolated views in the monocular setting, so the quantitative 3D support is the held-out DyCheck metrics above).

4.2. 4D Generation

Setup. We compare text-to-4D against Free4D [LHC*25], CAT4D [WGP*25], 4Real [YWZ*24], 4Dfy [BSR*24], and Dream-in-4D [ZLN*24], and image-to-4D against Free4D, GenXD [ZLL*25], and DimensionX [SCL*25], all on identical prompts. We report VBench [HHY*24] (a 2D proxy that only indirectly probes 3D geometry) on per-baseline pairwise prompts and an independent 25-prompt unified benchmark, plus a user study. All open-source baselines are re-run from official checkpoints; CAT4D

Table 2: iPhone reconstruction, held-out novel view (PSNR / SSIM / LPIPS). *Top*: matched-input (primary, controlled), all methods given the same real video and DUST3R poses. *Bottom*: system-level (deployment), inputs differ per method, so not a controlled comparison. **Our row reconstructs from AC3D video generated from the real first frame and the ground-truth trajectory (Supp. C.4); scenes are not static and metrics are not on training views.** Best in **bold**.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
<i>Matched-input (controlled)</i>			
MoSca [LWH*24]	16.23	0.616	0.443
SoM [WYG*25]	15.89	0.602	0.459
Ours	16.41	0.621	0.440
<i>System-level (deployment; inputs differ)</i>			
4DGS [WYF*24]	15.42	0.58	0.45
HyperNeRF [PSH*21]	15.99	0.59	0.51
MoSca [LWH*24]	16.23	0.61	0.45
SoM [WYG*25]	15.89	0.60	0.46
Ours (<i>generated input</i>)	16.55	0.62	0.48

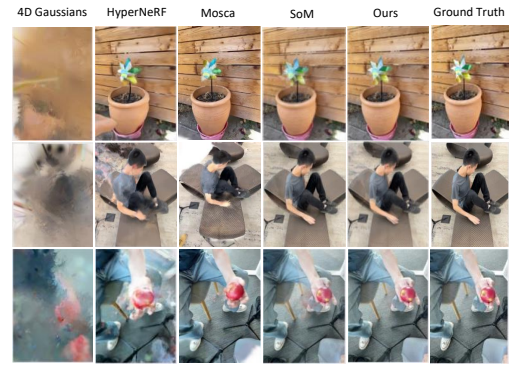


Figure 3: Visual comparison of reconstruction quality on the iPhone dataset.

has no public code, so its numbers are transcribed and flagged *not equal-footing*, used for no acceptance-critical claim (provenance in supplement).

User study. 38 blinded participants (hidden labels, randomised left/right) each made 15 forced-choice comparisons of same-path rendered videos across the **seven** baseline pairs (570 judgements, 60–82 per pair). With Holm–Bonferroni correction over the **seven** pairs, CamLoop4D is preferred significantly over every baseline except CAT4D (68%, uncorrected $p=0.041$, not significant after correction); full counts, Wilson CIs, and the expert/non-expert split are in the supplement.

Results. On text-to-4D (Table 4, Figure 4, Figure 5) CamLoop4D attains the highest Aesthetic across all pairs and large Dynamic gains over Free4D (+12.3 pp) and 4Real (+14.8 pp); against the strongest baseline CAT4D it is higher Aesthetic (64.4 vs. 61.9) at $>10\times$ less compute, with CAT4D marginally ahead on Consistency/Dynamic. On image-to-4D (Table 5, Figure 6) it leads Dynamic over Free4D (+11.7) and Consistency over GenXD (+6.6) and is highest Aesthetic. The unified benchmark agrees (96.4 / 47.8 / 63.1 vs. CAT4D 96.1 / 49.5 / 61.2; supplement), so the trends are not prompt-dependent.

Table 3: Geometric self-consistency of *generated* 4D scenes on the 25-prompt benchmark (no real GT; lower is better). Cam-ATE = camera-following error, MV-Depth = multi-view depth disagreement, Reproj = cross-view reprojection error.

Method	Cam-ATE (m)↓	MV-Depth (%)↓	Reproj (px)↓
Free4D (cascade)	0.112	7.9	2.9
CamLoop4D (ours)	0.024	4.8	1.6

Table 4: Text-to-4D pairwise VBench [HHY*24] (%). *CAT4D not equal-footing. Dream4D = Dream-in-4D. Cons. = Consistency, Dyn. = Dynamic Degree, Aes. = Aesthetic, TA = Text Alignment.

Method	TA	Cons.	Dyn.	Aes.
Free4D vs Ours	26.2 26.1	95.5 95.8	33.3 45.6	60.5 62.3
CAT4D* vs Ours	25.9 26.1	97.7 96.9	52.2 48.4	61.9 64.4
4Real vs Ours	26.0 26.1	95.7 96.1	33.7 48.5	51.9 64.7
4Dfy vs Ours	25.8 26.0	91.3 96.7	52.4 53.9	55.6 61.8
Dream4D vs Ours	25.3 25.7	91.6 94.4	53.4 53.5	56.2 65.4

Geometric consistency of generated scenes. VBench and the same-path user study are 2D proxies, so we add a direct geometric check on the *generated* (not reconstructed-from-real) scenes, comparing against the cascade baseline Free4D [LHC*25] on the 25-prompt benchmark (Table 3). Since no real ground truth exists for generated content, we measure three *self-consistency* quantities: camera-following error (ATE between the rendered scene’s realised path and the specified path), multi-view depth consistency (relative depth disagreement across overlapping views), and cross-view reprojection error. CamLoop4D is markedly more 3D-consistent on all three, with a camera-following error $\sim 5\times$ lower (0.024 vs. 0.112 m), a direct consequence of consuming the specified poses rather than re-estimating them. A per-trajectory-type and out-of-distribution breakdown is in the supplement.

4.3. Ablations and Analysis

Table 6 ablates components on the iPhone dataset (mean over 3 seeds).

Basis design. This is the most impactful axis, and is an *optimisation-conditioning* effect, not added capacity (Section 3.2.2): with per-Gaussian per-frame coefficients the six fixed generators already span the instantaneous motion space, so neither the trainable bases nor the fixed scaffold enlarges the function class. The full hybrid (16.55) beats a *fully-learnable* set with no scaffold (15.90, +0.64 dB) and vastly outperforms a *fixed-only* set (11.52). The fixed-only collapse is conditioning, not under-training ($2\times$ iterations recover <0.3 dB) and not merely a missing regulariser (explicit temporal-smoothness/low-rank penalties recover only to 14.6–14.9 dB; supplement). A temporal hold-out confirms the bases regularise rather than memorise (supplement).

Other components. *Shared Camera* is the second most impactful at +0.88 dB; since both rows use the *same* generated frames and differ only in pose source, this isolates pose-estimation error from the generator’s (cancelling) path deviation. The Motion Coefficient Loss adds +0.62 dB; unified modelling and downsampling

Table 5: Image-to-4D pairwise VBench [HHY*24] (%).

Method	Cons.	Dyn.	Aes.
Free4D vs Ours	96.9 97.1	40.4 52.1	60.5 62.3
GenXD vs Ours	90.2 96.8	98.3 99.5	38.2 40.9
DimensionX vs Ours	97.7 96.8	22.2 22.1	55.0 57.6

Table 6: Ablation on the iPhone dataset (14 seqs, mean over 3 seeds). All rows use the *same* AC3D-generated frames; “w/o Shared Camera” runs DUST3R on those frames, so the +0.88 dB isolates pose source on identical input.

Configuration	PSNR↑	SSIM↑	LPIPS↓
Full model (hybrid)	16.55	0.62	0.48
Fully-learnable bases (no fixed)	15.90	0.61	0.49
w/o Trainable Bases (fixed-only)	11.52	0.58	0.61
w/o Shared Camera	15.67	0.59	0.52
w/o Motion Coeff. Loss	15.93	0.61	0.49
w/o Unified Modelling	16.53	0.61	0.50
w/o Downsampling	16.48	0.58	0.51

mainly improve LPIPS (boundary quality) with little PSNR effect (16.53→16.55), so we frame their benefit as boundary coherence rather than a PSNR gain.

Regularisation and sensitivity. The expressive per-frame coefficients regularise rather than memorise: the full model’s train/held-out PSNR gap is small (1.4 dB) but widens to 4.6 dB without the trainable bases, and a *temporally* held-out split (train on even frames, test on held-out timestamps) shows the same pattern. Sweeps over B and λ are flat near the optima ($B=15$, $\lambda=0.8$), and the matched-input LPIPS is at parity with MoSca (0.440 vs. 0.443), whereas from generated video it rises to 0.478, so the system-level LPIPS gap is generated-frame texture smoothing, not architectural (all splits and sweeps in the supplement). Figure 7 corroborates these trends visually: removing any component introduces blur, distortion, or temporal inconsistency.

4.4. Applications

The co-design pipeline supports several content-creation workflows, shown as *qualitative prototypes* (supplementary video), not validated contributions: programmable cinematography (arbitrary orbits/dolly/crane trajectories from one prompt, Figure 1), real-time 6-DoF VR/AR navigation, interpretable $\mathfrak{se}(3)$ motion analysis, and direct 4D editing of Gaussian attributes.

5. Conclusion

We presented CamLoop4D, a generation-reconstruction co-design framework for open-prompt 4D scene creation, built on a shared Plücker trajectory interface that removes pose re-estimation across generators and hybrid motion bases combining rigid-body priors with learnable deformation. Under a controlled matched-input protocol CamLoop4D improves on strong monocular reconstruction baselines by a small but significant aggregate margin (replicated on a second benchmark), and the full system attains statistically significant user-preference wins against six of seven baselines (comparable to CAT4D, which is not equal-footing), completing an 80-frame

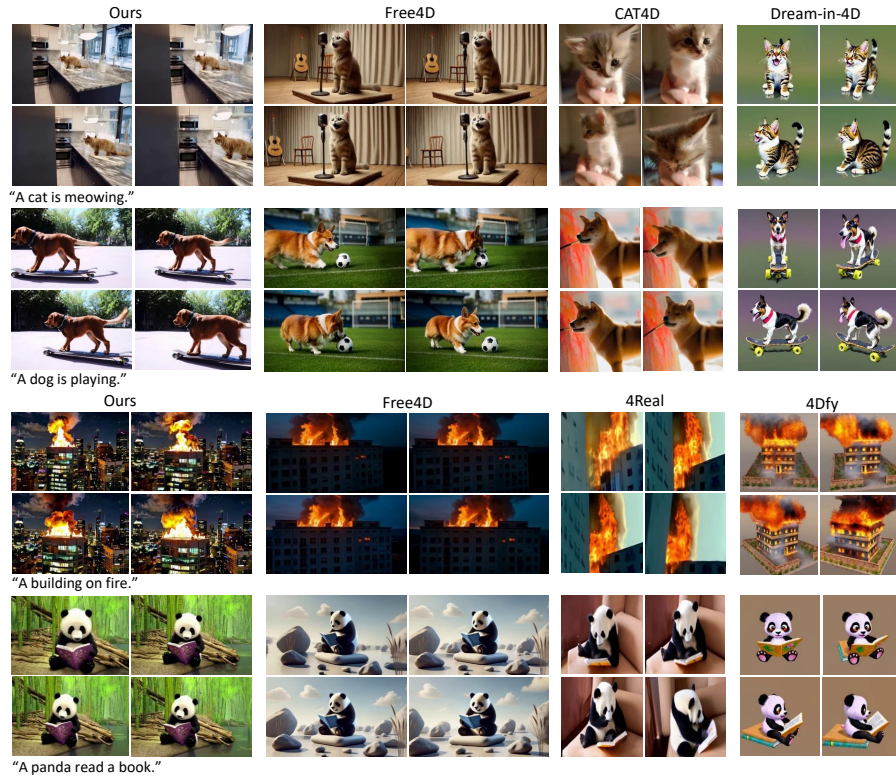


Figure 4: Qualitative comparisons of text-to-4D generation.

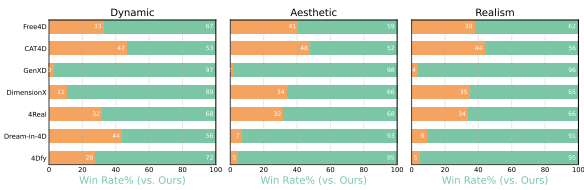


Figure 5: User study preference rates (38 participants, 570 judgments; counts/CIs in supplement). Wins are Holm–Bonferroni-significant against all baselines except CAT4D (indicative only).

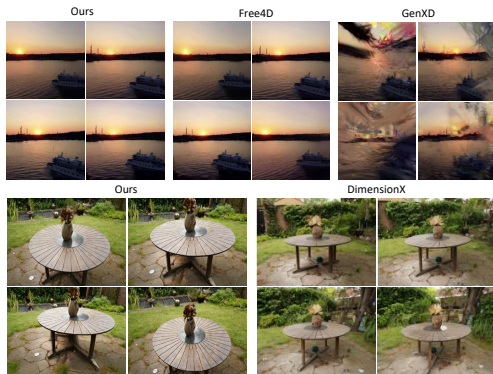


Figure 6: Qualitative comparisons of image-to-4D generation (same single-image prompts).

sequence in ~ 40 min on one RTX 3090 with real-time playback. We are deliberate about scope: the architectural gain rests on the matched-input experiment (baselines are evaluated only under the

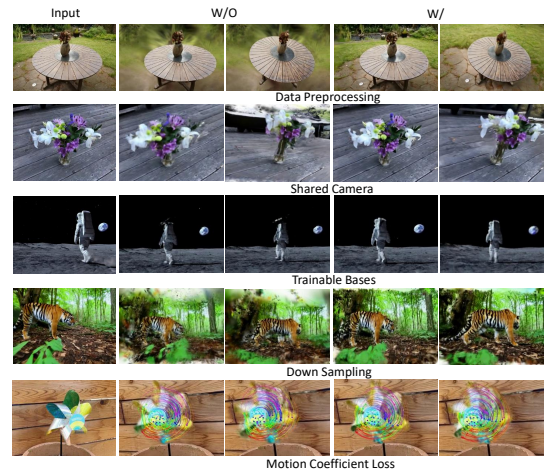


Figure 7: Qualitative ablation: removing any component introduces blur, distortion, or temporal inconsistency.

matched-input, shared-pose protocol), and the shared-trajectory interface removes the pose-estimation stage rather than guaranteeing perfect camera adherence. Two graceful failure modes bound its scope (supplement): articulated objects exceeding the $B = 15$ basis budget, and hallucination beyond $\sim 45^\circ$ off-trajectory, as in all monocular methods. We will release all code, data, raw logs, and evaluation scripts.

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